

IPOLIPID

1. Name of the medicinal product

Ipolid 300mg Capsule

2. Qualitative and quantitative composition

Each capsule contains 300 mg of gemfibrozil.

3. Pharmaceutical form

Capsule, hard

Description

Ipolid 300mg: maroon white hard gelatine capsule, size 0 printed 'IPOLIPID'

4. Clinical particulars

4.1 Therapeutic indications

Gemfibrozil is used in conjunction with dietary modification in the treatment of type IIa, type III and type V hyperlipoproteinemia. It is also indicated for the primary prevention of ischaemic heart disease in hyperlipidaemic patients who have not responded to dietary and other measures; in the USA this use is restricted to type IIb patients who also have low HDL cholesterol concentrations.

4.2 Posology and method of administration

Posology

Usual adult dose:

Orally 4 capsules (1.2g) a day in two divided dose thirty minutes before the morning and evening meals.

Usual paediatric dose:

Dose has not been established.

Method of administration

For oral use.

4.3 Contraindications

- Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.
- Hepatic impairment
- Severe renal impairment
- History of/or pre-existing gall bladder or biliary tract disease, including gallstones
- Concomitant use of repaglinide, dasabuvir, selexipag (see section 4.5), simvastatin, or rosuvastatin at 40 mg
- Patients with previous history of photo allergy or phototoxic reaction during treatment with fibrates

4.4 Special warnings and precautions for use

Muscle disorders (myopathy/rhabdomyolysis)

There have been reports of myositis, myopathy and markedly elevated creatine phosphokinase associated with gemfibrozil. Rhabdomyolysis has also been reported rarely.

Muscle damage must be considered in any patient presenting with diffuse myalgia, muscle tenderness and/or marked increase in muscle CPK levels ($>5\times$ ULN); under these conditions treatment must be discontinued.

Concomitant HMG CoA reductase inhibitors

The concomitant administration of gemfibrozil with simvastatin, as well as with rosuvastatin at 40 mg is contraindicated. Concomitant therapy of gemfibrozil with lower doses of rosuvastatin should be used only when the benefit outweighs the risks. There have been reports of severe myositis with markedly elevated creatine kinase and myoglobinuria (rhabdomyolysis) when gemfibrozil and HMG CoA reductase inhibitors were used concomitantly. Pharmacokinetic interactions may also be present and dosage adjustments may be necessary.

The benefit of further alterations in lipid levels by the combined use of gemfibrozil and HMG-CoA reductase inhibitors should be carefully weighed against the potential risks of such combinations and clinical monitoring is recommended.

A creatine phosphokinase (CPK) level should be measured before starting such a combination in patients with pre-disposing factors for rhabdomyolysis as follows:

- renal impairment
- hypothyroidism
- alcohol abuse
- age > 70 years
- personal or family history of hereditary muscular disorders
- previous history of muscular toxicity with another fibrate or HMG-CoA reductase inhibitor

In most subjects who have had an unsatisfactory lipid response to either drug alone, the possible benefits of combined therapy with HMG-CoA reductase inhibitors and gemfibrozil does not outweigh the risks of severe myopathy, rhabdomyolysis and acute renal failure.

Use in patients with gallstone formation

Gemfibrozil may increase cholesterol excretion into the bile raising the potential for gallstone formation. Cases of cholelithiasis have been reported with gemfibrozil therapy. If cholelithiasis is suspected, gallbladder studies are indicated. Gemfibrozil therapy should be discontinued if gallstones are found.

Monitoring serum lipids

Periodic determinations of serum lipids are necessary during treatment with gemfibrozil. Sometimes a paradoxical increase of (total and LDL) cholesterol can occur in patients with hypertriglyceridaemia. If the response is insufficient after 3 months of therapy at recommended doses treatment should be discontinued and alternative treatment methods considered.

Monitoring liver function

Elevated levels of ALAT, ASAT, alkaline phosphatase, LDH, CK and bilirubin have been reported. These are usually reversible when gemfibrozil is discontinued. Therefore, liver function tests should be performed periodically. Gemfibrozil therapy should be terminated if abnormalities persist.

Monitoring blood counts

Periodic blood count determinations are recommended during the first 12 months of gemfibrozil administration. Anaemia, leucopenia, thrombocytopenia, eosinophilia and bone marrow hypoplasia have been reported rarely.

Interactions with other medicinal products

Concomitant use with CYP2C8, CYP2C9, CYP2C19, CYP1A2, UGT1A1, UGT1A3 and OATP1B1 substrates

The interaction profile of gemfibrozil is complex resulting in increased exposure of many medicinal products if administered concomitantly with gemfibrozil.

Gemfibrozil potently inhibits CYP2C8, CYP2C9, CYP2C19, CYP1A2, and UDP glucuronyltransferase (UGT1A1 and UGT1A3) enzymes and also inhibits organic anion-transporting polypeptide 1B1 (OATP1B1). In addition, gemfibrozil is metabolised to gemfibrozil 1-O-β-glucuronide which also inhibits CYP2C8 and OATP1B1.

Concomitant use with hypoglycaemic agents

There have been reports of hypoglycaemic reactions after concomitant use with gemfibrozil and hypoglycaemic agents (oral agents and insulin). Monitoring of glucose levels is recommended.

Concomitant anticoagulants

Gemfibrozil may potentiate the effects of coumarin type vitamin K antagonist anticoagulants such as warfarin, acenocoumarol, or phenprocoumon. The concomitant administration of gemfibrozil with these anticoagulants necessitates careful monitoring of prothrombin time (INR - International Normalised Ratio). Caution should be exercised when such a coumarin type vitamin K antagonist anticoagulant is given concomitantly with gemfibrozil. The dosage of the anticoagulant may need to be reduced to maintain desired prothrombin time levels.

4.5 Interaction with other medicinal products and other forms of interaction

The interaction profile of gemfibrozil is complex. In vivo studies indicate that gemfibrozil and its metabolite gemfibrozil 1-O-β-glucuronide are potent inhibitors of CYP2C8 (an enzyme important for the metabolism of e.g., dabrafenib, enzalutamide, loperamide, montelukast, repaglinide, rosiglitazone, pioglitazone, dasabuvir, selexipag and paclitaxel). Co-administration of gemfibrozil with repaglinide, dasabuvir or selexipag is contraindicated. In

addition, dosing reduction of drugs that are mainly metabolised by CYP2C8 enzyme may be required when gemfibrozil is used concomitantly. In vitro studies have shown that gemfibrozil is a strong inhibitor of CYP2C9 (an enzyme involved in the metabolism of e.g. warfarin and glimepiride), but also of CYP 2C19, CYP1A2, OATP1B1 and UGT1A1 and UGT1A3. Gemfibrozil 1-O- β -glucuronide also inhibits OATP1B1.

Repaglinide

In healthy volunteers, co-administration with gemfibrozil increased the AUC and C_{max} of repaglinide by 8.1-fold and 2.4-fold, respectively. In the same study, co-administration with gemfibrozil and itraconazole increased the AUC and C_{max} of repaglinide by 19.4-fold and 2.8-fold, respectively. In addition, co-administration with gemfibrozil or with gemfibrozil and itraconazole prolonged its hypoglycaemic effects. Therefore, co-administration of gemfibrozil and repaglinide increases the risk for severe hypoglycaemia and is contraindicated.

Dasabuvir

Co-administration of gemfibrozil with dasabuvir increased dasabuvir AUC and C_{max} (ratios: 11.3 and 2.01, respectively) due to CYP2C8 inhibition. Increased dasabuvir exposure may increase the risk of QT prolongation, therefore, co-administration of gemfibrozil with dasabuvir is contraindicated.

Selexipag

Co-administration of gemfibrozil with selexipag, a substrate for CYP2C8, doubled exposure (AUC) to selexipag and increased exposure (AUC) to the active metabolite, ACT-333679, by approximately 11- fold. Concomitant administration of gemfibrozil with selexipag is contraindicated.

Enzalutamide

In healthy volunteers given a single 160 mg dose of enzalutamide after gemfibrozil 600 mg twice daily, the AUC of enzalutamide plus active metabolite (N-desmethyl enzalutamide) was increased by 2.2-fold and corresponding C_{max} was decreased by 16%. Increased enzalutamide exposure may increase the risk of seizures. Concomitant treatment of gemfibrozil and enzalutamide should be avoided; if co-administration is considered necessary, the dose of enzalutamide should be reduced.

Rosiglitazone

The combination of gemfibrozil with rosiglitazone should be approached with caution. Co-administration with rosiglitazone has resulted in 2.3-fold increase in rosiglitazone systemic exposure, probably by inhibition of the CYP2C8 isozyme.

HMG CoA reductase inhibitors

The concomitant administration of gemfibrozil with simvastatin, as well as with rosuvastatin at 40 mg is contraindicated. Concurrent use of fibrates with HMG-CoA reductase inhibitors may cause severe myositis and myoglobinuria. The combined use of gemfibrozil and a statin should generally be avoided. The use of fibrates alone is occasionally associated with myopathy. An increased risk of muscle related adverse events, including rhabdomyolysis, has been reported when fibrates are co-administered with statins.

Gemfibrozil has also been reported to influence the pharmacokinetics of simvastatin, lovastatin, pravastatin, rosuvastatin and atorvastatin. Gemfibrozil caused an almost 3-fold increased in AUC of simvastatin acid possibly due to inhibition of glucoronidation via UGT1A1 and UGT1A3, and a 3-fold increase in pravastatin AUC which may be due to interference with transport proteins. One study indicated that the co-administration of a single rosuvastatin dose of 80 mg to healthy volunteers on gemfibrozil (600 mg twice daily) resulted in a 2.2-fold increase in mean C_{max} and a 1.9-fold increase in mean AUC of rosuvastatin. The co-administration of a single lovastatin dose of 40 mg with gemfibrozil (600 mg twice daily for 3 days) in healthy volunteers resulted in a 2.8-fold increase of the mean AUC and C_{max} of lovastatin acid. The co-administration of a single atorvastatin dose of 40 mg with gemfibrozil (600 mg twice daily for 7 days) in healthy volunteers resulted in a 1.35-fold increase in mean AUC and no increase in mean C_{max} of atorvastatin.

Anticoagulants

Gemfibrozil may potentiate the effects of coumarin type vitamin K antagonist anticoagulants such as warfarin, acenocoumarol, or phenprocoumon. The concomitant administration of gemfibrozil with these anticoagulants necessitates careful monitoring of prothrombin time (INR).

Bexarotene

Concomitant administration of gemfibrozil with bexarotene is not recommended. A population analysis of plasma bexarotene concentrations in patients with cutaneous T-cell lymphoma (CTCL) indicated that concomitant administration of gemfibrozil resulted in substantial increases in plasma concentrations of bexarotene.

Bile acid – binding resins

Reduced bioavailability of gemfibrozil may result when given simultaneously with resin-granule drugs such as colestipol. Administration of the products two hours or more apart is recommended.

Colchicine

Risk of myopathy and rhabdomyolysis may be increased with concomitant administration of colchicine and gemfibrozil. This risk may be increased in the elderly and in patients with hepatic or renal dysfunction. Clinical and biological monitoring are recommended, especially at the start of combined treatment.

Gemfibrozil is highly bound to plasma proteins and there is potential for displacement interactions with other drugs.

4.6 Fertility, pregnancy and lactation

Pregnancy

There are no adequate data on use of gemfibrozil in pregnant women. Animal studies are insufficiently clear to allow conclusions to be drawn on pregnancy and foetal development. The potential risk for humans is unknown. Gemfibrozil should not be used during pregnancy unless it is clearly necessary.

Breast-feeding

There are no data on excretion of gemfibrozil in milk. Gemfibrozil should not be used when breast-feeding.

Fertility

Reversible decreases in male fertility have been observed in reproductive toxicity studies in rats

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. In isolated cases dizziness and visual disturbances can occur which may negatively influence driving.

4.8 Undesirable effects

Most commonly reported adverse reactions are of gastrointestinal character and are seen in approximately 7% of the patients. These adverse reactions do not usually lead to discontinuation of the treatment.

Adverse reactions are ranked according to frequency using the following convention: Very common ($\geq 1/10$), Common ($\geq 1/100$ to $< 1/10$), Uncommon ($\geq 1/1,000$ to $< 1/100$), Rare ($\geq 1/10,000$ to $< 1/1,000$), Very rare ($< 1/10,000$), including isolated reports:

System Organ Class	Undesirable effect
Blood and lymphatic system disorders	
Rare	Bone marrow failure, severe anaemia, thrombocytopenia, leukopenia, eosinophilia
Psychiatric disorders	
Rare	Depression, decreased libido
Nervous system disorders	
Common	Vertigo, headache
Rare	Neuropathy peripheral, paraesthesia, dizziness, somnolence
Eye disorders	
Rare	Vision blurred
Cardiac disorders	
Uncommon	Atrial fibrillation
Respiratory, thoracic and mediastinal disorders	
Rare	Laryngeal oedema
Gastrointestinal disorders	
Very common	Dyspepsia
Common	Diarrhoea, vomiting, nausea, abdominal pain constipation, flatulence
Rare	Pancreatitis, appendicitis

Hepatobiliary disorders	
Rare	Jaundice cholestatic, hepatitis, cholelithiasis, cholecystitis, hepatic function abnormal
Skin and subcutaneous tissue disorders	
Common	Eczema, rash
Rare	Angioedema, dermatitis exfoliative, urticaria, dermatitis, alopecia, photosensitivity reaction, pruritus
Musculoskeletal and connective tissue disorders	
Rare	Rhabdomyolysis, myopathy, myositis, muscular weakness, synovitis, myalgia, arthralgia, pain in extremity
Reproductive system and breast disorder	
Rare	Erectile dysfunction
General disorders and administration site conditions	
Common	Fatigue
Investigations	
Rare	Haemoglobin decreased, haematocrit decreased, white blood cell count decreased, blood creatine phosphokinase increased

4.9 Overdose

Overdose has been reported. Symptoms reported with overdosage were abdominal cramps, abnormal LFT's, diarrhoea, increased CPK, joint and muscle pain, nausea and vomiting. The patients fully recovered. Symptomatic supportive measures should be taken if overdose occurs.

5. Pharmacological properties

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Serum-lipid lowering agent

Chemical subgroup: Fibrates

ATC code: C10A B04

Gemfibrozil is a non-halogenated phenoxypentanoic acid. Gemfibrozil is a lipid regulating agent which regulates lipid fractions.

Gemfibrozil's mechanism of action has not been definitively established. In man, gemfibrozil stimulates the peripheral lipolysis of triglyceride rich lipoproteins such as VLDL and chylomicrons (by stimulation of LPL). Gemfibrozil also inhibits synthesis of VLDL in the liver. Gemfibrozil increases the HDL₂ and HDL₃ subfractions as well as apolipoprotein A-I and A-II.

Animal studies suggest that the turnover and removal of cholesterol from the liver is increased by gemfibrozil.

There is evidence that treatment with fibrates may reduce coronary heart disease events but they have not been shown to decrease all cause mortality in the primary or secondary prevention of cardiovascular disease.

5.2 Pharmacokinetic properties

Absorption

Gemfibrozil is well absorbed from the gastro-intestinal tract after oral administration with a bioavailability close to 100%. As the presence of food alters the bioavailability slightly gemfibrozil should be taken 30 minutes before a meal. Peak plasma levels occur in one to two hours. After administration of 600 mg twice daily a C_{max} in the range 15 to 25 mg/L is obtained.

Distribution

Volume of distribution at steady state is 9-13 L. The plasma protein binding of gemfibrozil and its main metabolite are at least 97%.

Biotransformation

Gemfibrozil undergoes oxidation of a ring methyl group to form successively a hydroxymethyl and a carboxyl metabolite (the main metabolite). This metabolite has a low activity compared to the mother compound gemfibrozil and an elimination half-life of approximately 20 hours. Glucuronidation to gemfibrozil 1-O-β-glucuronide is another important elimination pathway for gemfibrozil in man.

The enzymes involved in the metabolism of gemfibrozil are not known. The interaction profile of gemfibrozil and its metabolites is complex. In vitro and in vivo studies have shown that gemfibrozil inhibits CYP2C8, CYP2C9, CYP2C19, CYP1A2, UGT1A1, UGT1A3 and OATP1B1. Gemfibrozil 1-O-β-glucuronide also inhibits CYP2C8 and OATP1B1.

Elimination

Gemfibrozil is eliminated mainly by metabolism. Approximately 70% of the administered human dose is excreted in the urine, mainly as conjugates of gemfibrozil and its metabolites. Less than 6% of the dose is excreted unchanged in the urine. Six percent of the dose is found in faeces. The total clearance of gemfibrozil is in the range 100 to 160 ml/min, and the elimination half-life is in the range 1.3 to 1.5 hours. The pharmacokinetics is linear within the therapeutic dose range.

Special patient groups

No pharmacokinetic studies have been performed in patients with impaired hepatic function.

There are limited data on patients with mild, moderate and non-dialysed severe renal impairment. The limited data support the use of up to 1200 mg a day in patients with mild to moderate renal failure not receiving another lipid lowering drug.

6. Pharmaceutical particulars

6.1 List of excipients

Sodium starch glycolate

Magnesium stearate EP

Colloidal hydrated silica

Maize starch

6.2 Incompatibilities

Not applicable

6.3 Shelf life

5 years

6.4 Special precautions for storage

Store below 30°C in the original package

6.5 Nature and contents of container

PVC/PVDC-Alu blisters with 500 capsules

8. Manufacturer

Medochemie Ltd (AZ Factory), 2 Michael Erakleous Street, Agios Athanassios Industrial Area, Agios Athanassios, Limassol, Cyprus

9. Product Registration holder

Komedic Sdn Bhd, 4 Jalan PJS 11/14, Bandar Sunway 46150 Petaling Jaya, Selangor

10. Date of revision of the text

02/2023